What Does a Risk Guarantee Cost? Certified Hierarchical Back-off for Phenotype Normalization

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Abstract

Phenotype normalization systems map clinical mentions to concepts in the Human Phenotype Ontology (HPO), where top predictions are frequently incorrect, but broader ancestor concepts remain valid. We study whether a system can back off to such ancestors, or abstain, while certifying that its error rate on answered mentions stays below a target, which requires understanding and explicitly modeling the ontology's multi-inheritance directed acyclic graph (DAG) structure. We certify a fixed family of selective policies with Clopper-Pearson bounds and a Bonferroni correction. We then compare a learned policy with flat and climbing baselines. We report three findings. First, choosing thresholds by observed calibration risk exceeds the target in 40 of 69 cases, while certified selection never does. Second, on the standard PhenoNormBench evaluation, certified back-off reduces error from 17.2% to 0.9% and answers 70.9% of mentions, though climbing retains more topological information. Third, we show that grouping mentions by recorded dependencies makes informative certification difficult at strict risk targets: learned policies often abstain, while hierarchical climbing returns broader concepts. Certification is feasible under its sampling assumptions, but its practical utility depends on calibration quality, policy-family size, and the informativeness of permitted answers.

1 Introduction

The Human Phenotype Ontology (HPO) describes patient abnormalities with about 20,000 concepts arranged in a directed acyclic graph (DAG) (Gargano et al., 2024). Phenotype profiles coded in HPO support rare disease diagnosis, for example, through semantic similarity between patient and disease profiles (Köhler et al., 2009). Concept normalization systems map free-text mentions to HPO concepts (Luo et al., 2021; Groza et al., 2024; Alhassan et al., 2025). Embedding retrievers such as SapBERT (Liu et al., 2021) and BioLORD (Remy et al., 2024) are a common basis for this step.

These systems still make many errors. With BioLORD-based retrieval, 17.2% of top predictions on PhenoNormBench (Song et al., 2026) are neither a gold concept nor an ancestor of one. Such errors matter beyond normalization: unrelated phenotype annotations can distort similarity-based disease prioritization, while improved HPO curation has been shown to improve diagnostic ranking, boosting algorithmic accuracy from 66% to 86% in a study of systemic autoinflammatory diseases (Köhler et al., 2019; Maassen et al., 2023).

The ontology offers a natural alternative to a wrong answer. If a system is unsure whether a mention means *Atrial septal defect*, it can return the parent *Abnormal cardiac septum morphology*. The ancestor is less specific but still correct. Hierarchical selective prediction formalizes this trade-off between accuracy and specificity (Deng et al., 2012; Goren et al., 2024).

However, prominent existing formulations assume tree hierarchies with leaf-level ground truth and probability-based class scores. Phenotype normalization instead involves multi-inheritance DAGs, potentially internal gold concepts, and retrieval scores that need not be calibrated probabilities. Moreover, highly general ancestors may be hierarchically correct while providing little useful information.

These properties raise a central question: Can a phenotype normalizer guarantee that its error rate on answered mentions stays below a chosen target, and what does that guarantee cost in coverage and specificity?

We investigate this question through policies that retain exact predictions, back off to ancestors, or abstain. Our contributions are as follows.

- We formulate phenotype normalization as risk-controlled hierarchical selective prediction

over multi-inheritance DAGs with internal gold concepts, and apply family-wise finite-sample certification to a fixed set of policies (Section 3).

- We demonstrate the importance of certification: policies selected using observed calibration risk exceed the target in 40 of 69 evaluated cases (Section 6.1).

- We characterize the trade-off between certified risk, coverage, and specificity. On the standard benchmark, certified back-off answers 70.9% of mentions at a 5% risk target. Under stricter, dependence-aware partitions, we demonstrate that small sample sizes naturally restrict coverage, forcing methods to back off to broader concepts to satisfy the risk certificate (Sections 6.2 and 6.3).

- We compare learned back-off with hierarchical climbing, finding that neither consistently dominates: learned back-off achieves higher specificity at the cost of coverage, with gains at strict risk targets driven primarily by ancestor back-off rather than improved exact prediction (Sections 6.4).

2 Related Work

Phenotype concept normalization. Dictionary-based methods provide efficient phenotype recognition (Groza et al., 2024), while hybrid approaches combine lexical matching with neural models (Luo et al., 2021). More recent systems address discontinuous mentions through generative extraction and neural reranking (Alhassan et al., 2025), or integrate LLM-based extraction, HPO normalization, and phenotype prioritization into end-to-end pipelines (Shyr et al., 2026).

Embedding-based normalization builds on synonym-aware representations such as BioSyn and SapBERT (Sung et al., 2020; Liu et al., 2021), as well as ontology-grounded representations such as BioLORD (Remy et al., 2024). OntologyAligner combines ontology-aligned retrieval, large language model (LLM) based reranking, and hierarchy-guided refinement, and introduces PhenoNormBench (Song et al., 2026).

While these systems primarily optimize concept-level normalization accuracy, they do not establish finite-sample bounds on hierarchical error among returned predictions.

We instead keep the upstream normalizer fixed and study how specifically its predictions can be reported under a certified risk constraint.

Selective prediction. Selective classifiers abstain on uncertain inputs to reduce error (Chow, 1970; El-Yaniv and Wiener, 2010). Geifman and El-Yaniv (2017) calibrate confidence thresholds to control selective risk at a target level with high probability. Our Flat baseline follows the same keep-or-abstain decision structure, but uses our shared finite-family certification procedure.

Hierarchical prediction. Deng et al. (2012) back off to internal nodes of a class hierarchy to trade specificity for accuracy. Mortier et al. (2022) study hierarchical set-valued prediction, including representations that relax the restriction to sets encoded by a single internal node. Goren et al. (2024) extend selective classification to hierarchies and propose a climbing rule. Unlike their tree-based formulation with leaf-level ground truth and class probabilities, phenotype normalization involves multi-inheritance DAGs, potentially internal gold concepts, and ranked retrieval scores rather than a full class posterior. We therefore adapt climbing to Top-20 retrieval scores and the HPO graph, with explicit DAG-aware hierarchical validity.

Distribution-free risk control. Risk-controlling prediction sets calibrate set-valued outputs to control expected loss (Bates et al., 2021), while Learn then Test formulates risk control through multiple hypothesis testing over candidate parameters (Angelopoulos et al., 2025). Our certification step adopts this perspective using one-sided exact Clopper-Pearson binomial bounds (Clopper and Pearson, 1934) and a Bonferroni correction over a fixed, prespecified family of policies. The primary contribution of this step lies in the systematic integration of these distribution-free frameworks to compare learned and rule-based hierarchical policies under a common, non-asymptotic certification procedure over multi-inheritance ontology structures.

3 Task and Certification

Ontology. Let $\mathcal{H} = (V, E)$ be a DAG with root r and is-a edges. We use HPO release 2026-06-23 with $|V| = 19{,}836$ active concepts. For $v \in V$, $\mathrm{Anc}(v)$ denotes its strict ancestors and $\mathrm{Desc}^{+}(v)$ its descendants including v. We measure the *speci-*

ficity of a concept by its position in the graph:

$$\mathrm{spec}(v) = 1 - \frac{\log|\mathrm{Desc}^+(v)|}{\log|V|}. \quad (1)$$

It is 0 at the root and 1 at leaves.

Selective normalization. A mention m has a non-empty set of gold concepts $G(m) \subseteq V$. A frozen retriever returns $K = 20$ ranked candidates $c_1, \ldots, c_K$ with scores. A policy π sees only m and the candidates. It returns one of three actions: keep c_1 (KEEP), return a strict ancestor of c_1 (ANCESTOR), or abstain ($\perp$).

Risk. An answer $\hat{y} \neq \perp$ is *valid* if $\mathrm{Desc}^+(\hat{y}) \cap G(m) \neq \emptyset$, that is, if it is a gold concept or an ancestor of one. Otherwise it is an error. The *hierarchical risk* of π is its error rate on answered mentions:

$$R(\pi) = \Pr\big[\hat{y}_\pi \text{ is not valid} \mid \hat{y}_\pi \neq \perp\big]. \quad (2)$$

The root is always valid, so risk alone rewards uninformative answers. We therefore also report the *answer rate* $A(\pi)$, the *mean specificity* $S(\pi)$ of answers, and the *information coverage* $I(\pi) = A(\pi)\,S(\pi)$. Abstentions contribute zero to I. Specificity is a property of the graph, not a measure of clinical usefulness.

Certification. Let Π be a fixed family of M candidate policies, chosen before calibration labels are seen. On a calibration set, policy π answers n_π mentions and makes e_π errors. Its upper bound is the one-sided Clopper-Pearson limit at level δ/M:

$$U(\pi) = F^{-1}_{\mathrm{Beta}(e_\pi+1,\, n_\pi - e_\pi)}(1 - \delta/M), \quad (3)$$

with $U(\pi) = 1$ if $n_\pi = 0$ or $e_\pi = n_\pi$. A policy is *certified* at target ε if $U(\pi) \leq \varepsilon$. Each method j owns a subset $\Pi_j \subseteq \Pi$ and selects its certified policy with the most answers:

$$\pi_j^\star = \operatorname*{arg\,max}_{\pi \in \Pi_j :\, U(\pi) \leq \varepsilon} n_\pi. \quad (4)$$

If no policy in Π_j is certified, the method abstains on every mention.

Proposition 1. *Assume that (A1) the ontology and retriever are fixed, and the gold labels correctly specify hierarchical validity, (A2) calibration and test mentions are i.i.d. draws from the same distribution, and (A3) Π is fixed before calibration labels are seen. Then, with probability at least $1 - \delta$, $R(\pi) \leq U(\pi)$ for all $\pi \in \Pi$ simultaneously. Hence every selected $\pi_j^\star$ satisfies $R(\pi_j^\star) \leq \varepsilon$.*

The proof is a union bound over Π (Appendix A). The Bonferroni factor M is what makes the guarantee survive the search over policies. It also sets how much calibration data certification needs (Section 6.3). Assumption A2 is the one most at risk in practice, because benchmark mentions share patients, sources, and phrases. We test its effect directly in Section 6.3.

4 Policies

All evaluated policies share the underlying retriever, the top-20 candidate set, and the joint certification procedure. They differ exclusively in their structural decision rules for choosing among KEEP, ANCESTOR, and $\perp$. Figure 1 illustrates this overall framework, highlighting the multi-inheritance HPO DAG topology, the selective normalization decision space, and the operational boundaries between online inference and offline estimation.

Retrieval. We index the preferred label and every synonym of every active concept (44,814 strings). A frozen encoder f, either BioLORD-2023 or SapBERT, scores a concept by its best-matching string, $s(v) = \max_{j:\, c(j)=v} f(m)^\top f(t_j)$. We keep the top 20 concepts. No encoder is trained or fine-tuned.

Learned back-off (ours). The policy considers c_1 and its ancestors with specificity at least $\tau_s = 0.30$. The floor prevents near-root answers. For each such proposal a, we compute ten label-free features. They describe retrieval confidence (softmax mass over the Top-20 scores), agreement among candidates, the candidate mass under a, and the position of a in the graph. Two ℓ_2-regularized logistic regressions estimate $\hat{p}_e(m)$, the probability that c_1 is gold, and $\hat{p}_v(a \mid m)$, the probability that a is valid. In training, each mention's ancestors share a total weight of one, so long ancestor chains do not dominate. The policy keeps c_1 if $\hat{p}_e(m) \geq \tau_e$. Otherwise it returns the most specific ancestor with $\hat{p}_v(a \mid m) \geq \tau_a$, or abstains if there is none. The thresholds (τ_e, τ_a) range over a grid of at most 169 pairs built from fit data. The probabilities need not be calibrated, because certification applies to the resulting policy. Feature definitions and the grid are in Appendix B.

Baselines. **Raw Top-1** always returns c_1 and is not certified. **Flat** keeps c_1 or abstains and its decisions are made by thresholding a logistic correctness score using two features: the raw Top-1 retrieval score s_1 and its margin over the sec-

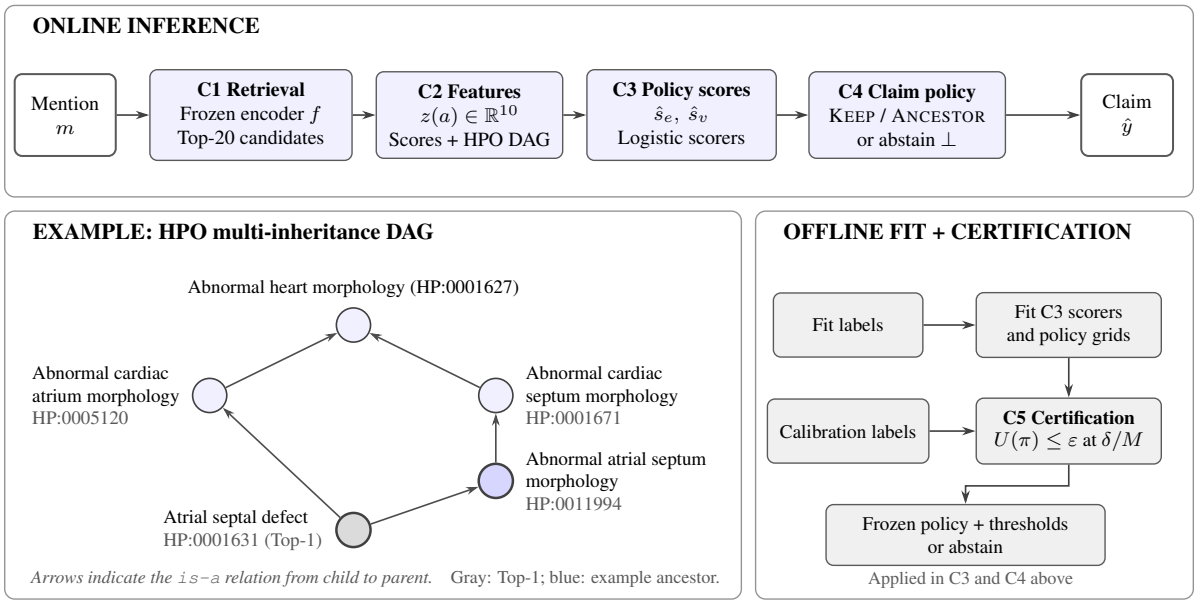


Figure 1: Overview of the risk-controlled hierarchical normalization framework. Online inference retrieves HPO candidates (C1), computes retrieval and ontology features (C2), scores candidate claims (C3), and selects an exact concept, an ancestor, or abstention (C4). The HPO example illustrates multiple inheritance, with directed edges from child concepts to their parents. Offline fitting and calibration (C5) select a policy under a finite-family risk bound.

ond candidate $(s_1 - s_2)$ (Geifman and El-Yaniv, 2017). **Hierarchy-only** shares Flat's Top-1 correctness model but backs off using the candidate mass under each ancestor, without a separately learned ancestor-validity model. **Ancestor-only** is our policy without the KEEP action. **Climbing** (Goren et al., 2024) moves from c_1 toward the root until the candidate mass under the current node passes a threshold. Hierarchical selective classification (HSC) assumes a tree and class probabilities, so we use two adaptations to the HPO graph: a tree projection and a direct DAG version. A third variant applies a specificity floor of 0.30 to the DAG adaptation. These are adaptations, not replications. **Empirical tuning** uses our policy family but selects the policy with the most answers whose *observed* calibration risk e_π/n_π is at most ε.

Policy family. The family Π contains every threshold setting of the seven certified rule families (ours, Flat, Hierarchy-only, Ancestor-only, and three Climbing variants) for both encoders, for a total of $M = 1{,}279$ policies. All methods share this M and $\delta = 0.05$.

5 Experimental Setup

Data. PhenoNormBench (Song et al., 2026) contains 13,390 phrase and HPO ID pairs from seven corpora, including accepted alternative IDs. Among rows with a single gold concept, 54.5% map to non-leaf concepts, so back-off often has room to be correct. From the 2,100-row development release, we deduplicate phrases (1,514 rows) and split them into a fit set (500), a calibration set (500), and a development test set (514).

Corpus-level counts and the subset without recorded development overlap are detailed in Table 4.

Standard evaluation. We freeze the BioLORD policies at $\varepsilon = 0.05$ and evaluate them on the remaining 11,290 rows without recalibration. Many of these rows share phrases or patients with the development data. This evaluation does not establish the i.i.d. assumption A2 required for the population-risk guarantee.

Dependence-aware evaluation. To approximate A2, we build connected components over all 13,390 rows, linking rows that share a patient or article identifier or a normalized phrase. Components never cross fit, calibration, and test. Calibration and test sets use one mention per component. We repeat this over 20 predefined splits and two fit populations, refitting all models and fixing the policy family before calibration each time. This design controls only the dependencies recorded in the

metadata. Details are in Appendix C.

Our matched comparison evaluates four policy families—learned back-off, Hierarchy-only, and two DAG Climbing variants—using a restricted joint Bonferroni budget ($M = 409$–422) based on prespecified parameter grids. Learned back-off and Hierarchy-only share identical retrieval configurations, strictly isolating the effect of learned ancestor scoring. This controlled setup is distinct from the unconstrained global evaluation ($M = 1{,}279$). Grouping controls only recorded metadata dependencies, and all comparisons remain retrospective; full split and policy details are in Appendix C.

Transfer. We also apply the method to candidates from OntologyAligner (514 rows) and to BioCreative V Chemical–Disease Relation (BC5CDR) (Li et al., 2016), whose disease mentions are linked to MEDIC (Davis et al., 2012), a second DAG. For BC5CDR, we use the official article-level splits and sample one mention per article for calibration and evaluation.

Metrics and statistics. We report answer rate, hierarchical risk, mean specificity, information coverage, and the share of mentions on which the policy keeps c_1. Intervals are 95% percentile intervals from 2,000 cluster-bootstrap resamples (Efron, 1979). Pairwise tests use 10,000 cluster-level paired permutations with Holm correction (Holm, 1979). Only the frozen operating point in Table 1 was fixed before its evaluation labels were seen. All other analyses reuse inspected data and are descriptive.

6 Results

Unless noted, results use BioLORD and $\varepsilon = 0.05$.

6.1 Observed calibration risk is not enough

Empirical tuning exceeds the target on the test set in 40 of its 69 nonempty selections. Certified selection exceeds it in 0 of 40 (Figure 2). Empirical tuning overfits to calibration noise, especially when many policies are compared. Certification has a price. For our policy, it lowers the answer rate from 83.7% to 69.6%.

6.2 Standard evaluation: back-off lowers risk, climbing keeps more information

Raw Top-1 errs on 17.2% of mentions (Table 1). At a 5% target, our policy answers 70.9% of mentions with 0.85% observed risk. Almost all of its

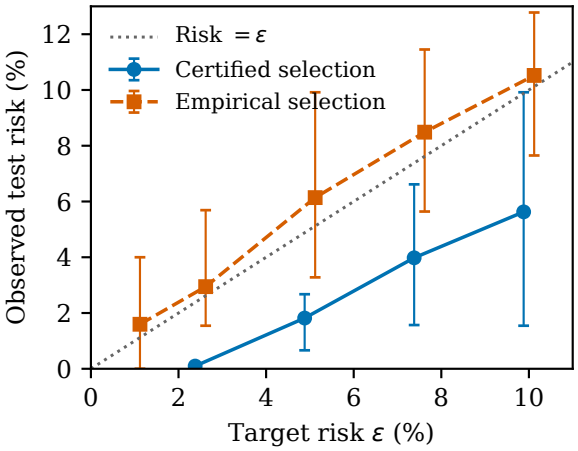


Figure 2: Observed test risk of the selected policy against the target risk, on the 514 development test rows. Points are means over policy families and encoders. Whiskers show ranges. No policy is certified at $\varepsilon = 1\%$. These retrospective test comparisons are descriptive, not validation of population-risk guarantees.

answers are ancestors. It keeps c_1 on only 0.5% of rows. It abstains on 3,291 rows, 2,002 of which had a correct c_1. Flat and Hierarchy-only certify no policy at this target.

Climbing answers every mention with about 2% risk. Its historical answers are less specific (52.5 and 56.2 versus 60.2), but because it never abstains, it has higher information coverage than ours (52.5 and 56.2 versus 42.7; Holm-adjusted $p = 0.0011$ for the frozen comparisons).

The full frontier confirms this pattern (Figure 3). No method certifies at 1%. At 2.5%, only Climbing certifies, and it does so by returning near-root concepts (specificity 5.9 and 12.0). At 7.5% and 10%, our policy keeps c_1 more often (64.9% and 56.4% of rows), but Climbing still has slightly higher information coverage. Averaged over the five targets, information coverage is 38.7% for ours, 44.3% for tree Climbing, and 48.1% for DAG Climbing. The two approaches make different trade-offs. Ours gives more specific answers and abstains more. Climbing always answers but broadens more.

6.3 Dependence-aware evaluation: the certificate becomes uninformative

When dependent mentions are kept out of calibration, the picture changes (Table 2). At a 5% target, our policy is certified in only 1 and 4 of 20 splits, so it answers 3.0% and 11.7% of mentions. Climbing is certified in every split, but it reaches the target by returning broad concepts (specificity 13.5 to 19.3). At this target, learned back-off rarely

Table 1: Standard evaluation on 11,290 PhenoNormBench rows (BioLORD, $\varepsilon = 5\%$; all values in %). Risk is the error rate on answered mentions, with 95% cluster-bootstrap intervals. Keep: mentions on which the policy returns c_1. "none": no policy certified, so the method abstains on all rows. The DAG row reports the frozen historical outputs (56.2% information coverage). These evaluation rows overlap with development data, so the population certificate does not formally apply.

Method	Answer	Risk	Spec.	Info.	Keep
Raw Top-1 (not certified)	100.0	17.19 [16.21, 18.41]	86.9	86.9	100.0
Flat		none			
Hierarchy-only		none			
Climbing (DAG, floor)		none			
Climbing (tree)	100.0	1.94 [1.57, 2.44]	52.5	52.5	34.9
Climbing (DAG)	100.0	2.10 [1.69, 2.62]	56.2	56.2	37.0
Ancestor-only	70.4	0.78 [0.53, 1.10]	60.5	42.6	0.0
Learned back-off (ours)	70.9	0.85 [0.59, 1.19]	60.2	42.7	0.5

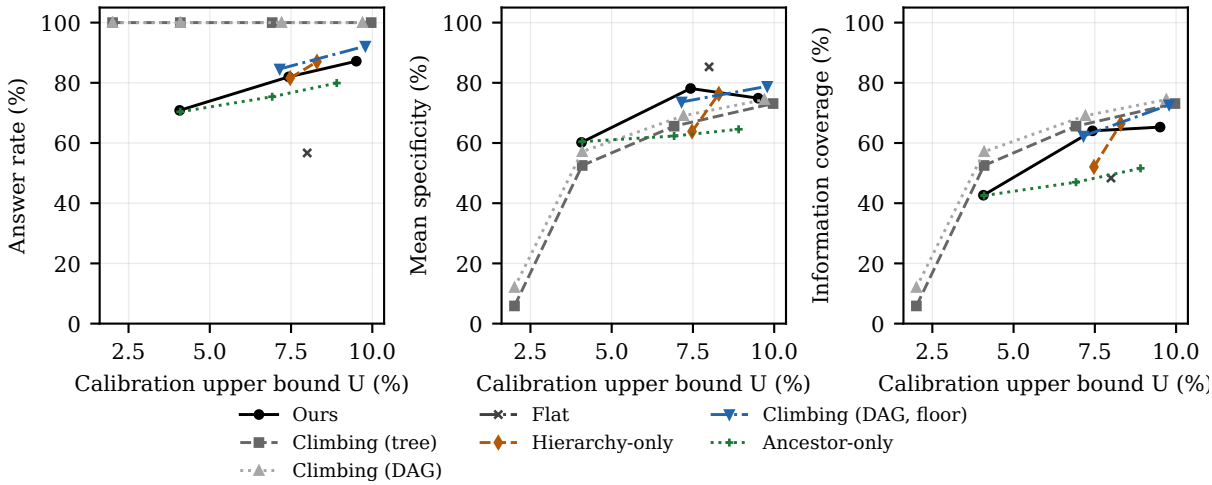


Figure 3: Certified operating points on the standard evaluation for targets $\varepsilon \in \{1, 2.5, 5, 7.5, 10\}\%$, plotted against the calibration upper bound U. Only nonempty selections are shown. No method is certified at 1%. Additional numerical results are reported in Appendix D.

certifies, whereas unrestricted Climbing retains information by returning substantially broader concepts. At 10%, our policy certifies in 19 and 20 of 20 splits with information coverage of 52.7% and 57.3%, against 58.5% to 61.9% for Climbing (Appendix D).

Calibration size helps explain these failures. With $M = 1{,}279$, our policy certifies at 5% only when calibration has at least about 400 rows (Figure 4b). The grouped protocols each use 500 calibration components, however; certification depends on the number of answered calibration cases, their errors, and the multiplicity correction, not calibration size alone. For example, with $M = 1{,}279$ and $\delta = \varepsilon = 0.05$, a policy with zero calibration errors requires at least 198 answered calibration examples; with five errors, it requires at least 408. The observed onset near 400 rows is therefore protocol-specific. The grouped test sets contain 818 or 518 representative components, depending on the fit population (Appendix C). The multiplicity correction, and thus the size of the policy family, largely determines how much calibration data certification needs.

6.4 What the learned components contribute

At the 5% target, the gains of our policy come from ancestor back-off. Ancestor-only, which never keeps c_1, matches it almost exactly (Table 1). Removing the KEEP model changes nothing on the development set (Table 3). Removing ancestor back-off, the ancestor model, or the graph features prevents certification at 5%. Removing retrieval-score features or sample weights keeps certification but lowers information coverage.

The KEEP model matters at looser targets. There, our policy keeps c_1 on more than half of the rows, and its average information coverage across tar-

Table 2: Dependence-aware evaluation at $\varepsilon = 5\%$, over 20 splits per fit population (%). Cert./20: splits in which a policy was certified. Answer and Info. count uncertified splits as full abstention. Risk and Spec. average over splits with answers.

Fit population	Method	ε	Cert./20	Answer	Risk	Spec.	Info.
Large component	Ours	5	1	3.0	2.7	60.7	1.8
Large component	Climbing (tree)	5	20	100.0	0.6	15.5	15.5
Large component	Climbing (DAG)	5	20	100.0	0.5	19.3	19.3
Small components	Ours	5	4	11.7	2.6	63.6	7.5
Small components	Climbing (tree)	5	20	100.0	0.4	13.5	13.5
Small components	Climbing (DAG)	5	20	100.0	0.4	18.7	18.7

Table 3: Component ablation on the 514 development test rows (BioLORD, $\varepsilon = 5\%$; %). "–": no policy certified. Min. ε: smallest certified target in $\{1, 2.5, 5, 7.5, 10\}$.

Variant	Ans.	Risk	Info.	Min. ε
Full (ours)	69.6	1.1	42.7	5
w/o KEEP model	69.6	1.1	42.7	5
w/o sample weights	68.3	0.9	40.7	5
w/o score features	54.5	1.1	29.1	5
w/o graph features	–	–	–	7.5
w/o ancestor model	–	–	–	7.5
w/o ancestor back-off	–	–	–	10

gets exceeds that of Ancestor-only (38.7% versus 32.0%). The learned models are not always necessary. With SapBERT, Hierarchy-only certifies at 5% and reaches higher information coverage than ours (Appendix D).

While the learned policy provides more specific and topologically informative answers on the standard evaluation, it does so at a substantial coverage cost under strict partitions. At the strict grouped target, unrestricted DAG Climbing certifies more consistently and retains higher overall information coverage. These trade-offs indicate that a more complex parametric model does not consistently dominate a simple hierarchical rule across the evaluated settings.

In the matched grouped comparison, learned back-off certifies in 1 versus 5 of 20 splits for Hierarchy-only under large-component fitting, and in 4 versus 3 under small-component fitting (Table 5). Learned ancestor scoring therefore does not consistently improve certification or information coverage over structural branch mass.

One possible explanation is that the retrieval mass under each ancestor already captures much of the hierarchical evidence used by the learned model. Additional learned features may provide limited benefit when fitting data are small or dependent, particularly because certification depends on calibration errors rather than predictive scores alone. These explanations remain hypotheses rather than established causes of the observed differences across encoders.

6.5 Transfer

With OntologyAligner candidates, tree Climbing certifies only a near-root operating point. Our policy answers 82.7% of mentions at 3.5% observed risk and specificity 0.78, against 11.1% risk for Raw Top-1. On BC5CDR, our policy answers 266 of 493 mentions without error (information coverage 29%). Tree Climbing answers all mentions with 1.2% risk (35%). Both transfer studies are retrospective, so we treat them as descriptive.

7 Error Analysis

On the standard evaluation, our policy makes 68 errors among 7,999 answered mentions (0.85%). All occur when c_1 is wrong. On rows with a wrong c_1, the policy answers 44.0% and errs on 6.7% of those answers. The specificity floor applies only to ancestor proposals, preserving the option to return the retriever's Top-1 concept. All six observed KEEP errors involved broad concepts with specificity at most 0.19. Applying the floor to both actions would exclude these broad KEEP outputs, but its effects on coverage, risk, and certification remain untested. Errors concentrate in corpora with repeated mentions. They include over-specific answers, answers in a different branch, near misses, and possible annotation mismatches. We count all 68 as errors under the official gold labels. Mentions of five or more words have lower coverage and higher risk (Appendix E).

8 Conclusion

We studied whether phenotype normalization can come with a guaranteed error rate on answered

mentions. In our retrospective comparisons, certified selection avoids the observed test-risk exceedances associated with empirical tuning. On the standard benchmark, certified back-off reduces observed error from 17.2% to under 1% while answering most mentions. Under dependence-aware splits, learned back-off rarely certifies at a 5% target, whereas Climbing retains information by returning broader concepts. Learned back-off and hierarchical climbing make different trade-offs, and neither dominates. For practitioners, the main levers are the amount of independent calibration data, policy-family size, and the specificity of permitted answers. Future work should test smaller, prespecified policy families and collect calibration data from independent sources.

Limitations

All evaluations are retrospective. Only the frozen operating point on the standard benchmark was fixed before its labels were seen, and that benchmark overlaps with the calibration data. Certification across grouped splits does not establish robustness to unobserved dependence or distribution shift. Clinical deployment would require prospective validation and calibration using appropriately independent sampling units, potentially with patient- or institution-level grouping. The guarantee assumes correct gold labels, but some errors may be annotation mismatches. Specificity is a graph property, so information coverage does not measure clinical usefulness, and we did not run a clinical or human evaluation. The Bonferroni correction is conservative for large policy families. Climbing was adapted to retrieval scores and the HPO graph, so our results may not reflect its original setting. Most results use one ontology release and one main benchmark. Future work should assess the clinical informativeness of ancestor predictions through blinded expert ratings of their diagnostic relevance and usefulness in patient cases. Such evaluations could distinguish topological specificity from information that actually supports clinical reasoning.

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A Proof of Proposition 1

Fix a policy π and let B_i indicate whether it answers calibration mention i, and E_i indicate an invalid answer. Under A1–A3, the calibration examples are i.i.d., and the policy and its family are fixed independently of their labels. If $\Pr(B_i = 1) > 0$, conditional on $n_\pi = \sum_i B_i > 0$, the number of errors satisfies $e_\pi \sim \mathrm{Binomial}(n_\pi, R(\pi))$. Thus the one-sided Clopper–Pearson bound gives $\Pr[R(\pi) > U(\pi) \mid n_\pi] \le \delta/M$. If $n_\pi = 0$, we set $U(\pi) = 1$ and the policy cannot certify for $\varepsilon < 1$; if its population answer probability is zero, conditional risk is undefined and the policy is never selected. Averaging over n_π and taking a union bound over the fixed family yields

$$\Pr[\exists \pi \in \Pi :\ R(\pi) > U(\pi)] \le \delta. \qquad (5)$$

Hence every selected $\pi_j^\star$ with $U(\pi_j^\star) \le \varepsilon$ satisfies $R(\pi_j^\star) \le \varepsilon$ on the same event. The union bound does not require policies to be independent. With no observed errors, certification requires $n_\pi \ge \lceil \log(\delta/M)/\log(1-\varepsilon) \rceil$ answered calibration examples.

B Policy Details

Ontology and retrieval. We use HPO release 2026-06-23 (19,836 active terms and 24,378 `is_a` edges). Retrieval indexes 44,814 preferred-label/synonym strings and assigns each concept its highest cosine similarity across indexed strings. The frozen BioLORD-2023 and SapBERT encoders return the top 20 concepts. Alternative IDs and unambiguous `replaced_by` chains are resolved; unresolved IDs cannot establish validity.

Features and fitting. Write $p_k = \exp(s_k/T_0)/\sum_{j\le 20}\exp(s_j/T_0)$, where $T_0 = 0.05$, and $\mu(a) = \sum_{k:c_k\in \mathrm{Desc}^+(a)} p_k$. The ten features are s_1, s_2, $s_1 - s_2$, entropy $-\sum_k p_k \log p_k$, $\mu(a)$, longest-path depth, strict descendant count, $\mathrm{spec}(a)$, common-ancestor count across the retrieved concepts, and the number of retrieved concepts below a. Each concept contributes only once to $\mu(a)$, even with multiple inheritance. Separate standardized, ℓ_2-regularized logistic models score Top-1 correctness and ancestor validity ($C = 1$, L-BFGS, 2,000 iterations). Ancestor examples are weighted so

Corpus	All	Dev.	Clean
GeneReviews-10	352	300	0
ID-68	866	300	11
GSC2017	2,773	300	203
GSC2024	2,910	300	213
CSC	1,783	300	155
FGDD	1,866	300	316
BC8-T3	2,840	300	1,697
Total	13,390	2,100	2,595

Table 4: PhenoNormBench corpus sizes. "Clean" denotes evaluation rows without recorded normalized-phrase or patient overlap with development; it does not imply independent sampling.

that each mention contributes total weight one. These outputs are scoring functions, not calibrated probabilities.

Selection and baselines. Fit-derived quantiles and boundary thresholds generate at most 169 learned policy configurations. Among certified configurations, selection maximizes calibration answers, then specificity, then the number of KEEP actions, then uses the configuration index. Flat thresholds a two-feature logistic Top-1 score $(s_1, s_1 - s_2)$. Hierarchy-only uses the same Top-1 scorer but replaces learned ancestor scores by $\mu(a)$; Ancestor-only disables KEEP. Climbing uses Top-20 score mass, either on a fixed tree projection or directly on the HPO DAG; the third DAG variant excludes strict ancestors with specificity below 0.30. The tree projection chooses the deepest parent (then HPO ID); DAG descendant mass counts each retrieved candidate once. These HSC implementations adapt, rather than replicate, the original tree/posterior setting. The historical joint comparison uses $M = 1{,}279$ and $\delta = 0.05$.

C Setup Details

Standard split and recorded overlap. Phrases are stripped, case-folded and whitespace-normalized. The 2,100 development rows are shuffled with seed 2026 and deduplicated by phrase, then reshuffled with seed 2027 and partitioned into 500 fit, 500 calibration and 514 descriptive rows. The other 11,290 rows are evaluated without recalibration; 4,687 share a phrase and 7,101 share a known patient with development (overlapping counts). Only 2,595 share neither recorded identifier (Table 4).

Dependence-aware splits and matched comparison. Connected components join normalized phrases and known patient/article identifiers (corpus-local namespaces except across GSC releases). There are 1,319 components, one containing 11,351 rows (84.8%). For seeds 2026–2045, fitting uses either 500 rows from the largest component or representatives of 300 small components; calibration uses 500 disjoint small-component representatives and testing the remaining 818 or 518. The matched comparison fixes four families before calibration: learned (169 settings), Hierarchy-only with shared KEEP scorer (156–169), DAG Climbing with a specificity floor (42), and unrestricted DAG Climbing (42), for joint $M = 409$–422. The learned and Hierarchy-only families share inputs and ancestor proposals; only the ancestor scoring differs. Corrected DAG Climbing rounds masses to 12 decimals before eligibility and tie-breaking by specificity and HPO ID. These results are retrospective and separate from the historical $M = 1{,}279$ comparison.

Transfer, computation and uncertainty. OntologyAligner uses exported candidate rankings; its native "No Match" entails abstention. BC5CDR follows the official article splits, one representative mention per article (seed 2026), and MEDIC with a synthetic root; the reported evaluation contains 493 test articles with resolvable gold IDs. Unresolved labels and unknown overlap limit transfer conclusions. No neural encoder is fine-tuned and no LLM is called by our policy experiments. Standard confidence intervals use 2,000 patient/phrase-cluster bootstrap resamples; paired comparisons use 10,000 cluster-level permutations with Holm correction. Exact software versions, settings and hashes are provided in the reproducibility artifacts.

D Additional Results

Matched grouped comparison. Table 5 compares four policy families on identical component memberships under a shared joint Bonferroni correction.

Encoder, transfer and overlap checks. With SapBERT, Hierarchy-only certifies at 5% and retains more information than learned backoff; encoder-specific comparisons remain descriptive. On OntologyAligner candidates (514 rows), Learned answers 82.7% with 3.5% observed risk and 64.9% information coverage; tree Climbing answers all rows at 0.4% risk and 7.1% information coverage. On BC5CDR (493 resolved-gold arti-

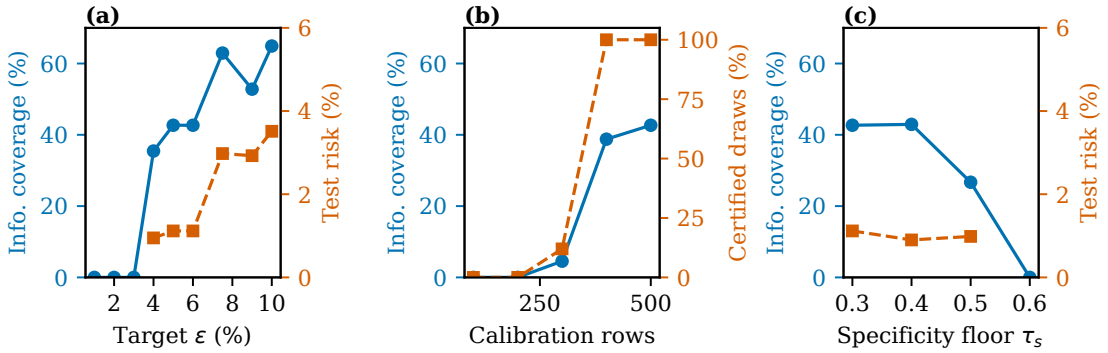


Figure 4: Development-set sensitivity: (a) information coverage and observed risk across targets; (b) information coverage and certification frequency by calibration size at $\varepsilon = 5\%$; (c) sensitivity to the ancestor-specificity floor. All results are retrospective.

Table 5: Matched dependence-aware comparison at $\varepsilon = 5\%$ under joint Bonferroni correction ($M = 409$–422). Cert.: certified splits out of 20; Info.: information coverage (%), including uncertified splits as zero.

Fit set	Policy	Cert.	Info.
Large	Learned	1/20	2.0
	Hierarchy-only	5/20	9.9
	DAG Climbing (floor)	4/20	6.5
	DAG Climbing (free)	20/20	34.8
Small	Learned	4/20	8.2
	Hierarchy-only	3/20	6.1
	DAG Climbing (floor)	4/20	6.4
	DAG Climbing (free)	20/20	34.7

cles), Learned answers 54.0% without an observed error and retains 29.3% information coverage; tree Climbing answers all rows at 1.2% risk and 34.6% information coverage. On the 2,595 rows free of *recorded* development overlap, the frozen learned policy has 1.32% observed risk (95% interval [0.76, 2.04]). None of these transfer or subset analyses establishes the sampling assumptions for certification.

Historical DAG audit and grouped frontier. The historical frozen DAG Climbing predictions retain 56.2% information coverage at 5% risk target; a subsequent mass-tie audit yields 57.1% at the same threshold and 2.10% observed risk. These are not the same prediction files, and the historical significance test is not a test of the corrected variant. In the historical grouped 10% analysis, Learned certifies in 19/20 and 20/20 splits (52.7% and 57.3% information coverage), compared with 58.5–61.9% for Climbing.

E Additional Analysis

Error patterns. Among 7,999 answered mentions, 68 predictions are invalid under the official gold labels. The six erroneous KEEP outputs correspond to broad concepts (specificity ≤ 0.19); this occurs because the underlying retriever returned a general term as the top candidate c_1, which the policy retained since the specificity floor applies exclusively to ancestor proposals. Observed test risk is 10.9% for KEEP actions and 0.78% for ancestor answers, noting that these action-specific quantities are not individually certified. Qualitative error analysis identifies over-specific predictions, different-branch answers, near misses, and possible annotation mismatches. Finally, longer mentions (five or more words) exhibit lower answer coverage and higher observed risk.